

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2022-23

Paper: GLB-502 — Igneous Petrology, Mineralogy and Crystallography

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 052658

*Instructions: Answer **five** questions in all, selecting at least **two** questions each from Sections A and B. Section-C is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. Justify the following sentences (if necessary with suitable sketches): [14]

- (a) The solvus in the orthoclase-albite system is convex upward.
- (b) Calc-alkaline rocks are confined to subduction zones.
- (c) Pegmatite and aplite have the same mineralogical composition as granite but have different textures.
- (d) Fractional crystallization does not affect the path followed by the liquid in eutectic systems.

Question 2. Discuss the following two processes using any suitable phase diagram: [14]

- (a) The formation of resorption texture.
- (b) Normal zoning in plagioclase.

Question 3. Differentiate between **ANY THREE** of the following: [14]

- (a) Spherulitic growth and variolitic growth
- (b) Homogeneous and heterogeneous nucleation
- (c) Continuous and discontinuous reaction
- (d) Orthocumulate and adcumulate

Question 4. Answer the following parts: [14]

- (a) A geologist working in cratonic terrain wanted to classify the rock types of the terrain, and he noted the following observations for rocks at distant places:

Medium to coarse-grained rocks, dominated by felsic minerals, presence of the following number of minerals after counting many observation grids:

- (i) Quartz 30; alkali feldspar 60; plagioclase feldspar 10; biotite 07; tourmaline 03.

- (ii) Quartz 11; alkali feldspar 39; plagioclase feldspar 85; Mica 5.

Name Rock I and Rock II as per the above description. [5]

- (b) Draw the IUGS classification diagram for plutonic ultramafic rock types. [5]

- (c) Describe the Dacite and Granodiorite rock types. [4]

SECTION — B

Question 5. Write an essay on the mica group of minerals giving the general formula, structure, classification, and paragenesis with neat sketches, where applicable. [14]

Question 6. Write short notes on **ANY TWO** of the following: [14]

- (a) Classification of feldspars
- (b) H-M Symbols (Hermann-Mauguin symbols)
- (c) Factors controlling solid solution in minerals

Question 7. How do we project any crystal or crystal model from spherical projection to stereographic projection? Describe the details of the stereonet which is used in crystallography and mention various symbols used to describe symmetry operations. Write a step-wise description of plotting φ on the stereonet. [14]

SECTION — C

Question 8. Answer the following in one word or sentence: [10 × 1.4 = 14]

- (a) What is the difference between zoning and twinning in minerals?
- (b) What is the significance of inverted pigeonite?
- (c) What is the importance of Ringwoodite?
- (d) Three 2-fold axes each perpendicular to a mirror plane. Each 2-fold axis is coincident with a crystallographic axis. Identify the crystal class, class name, and at least one example.
- (e) Cross-hatched appearance in microcline results due to _____.
- (f) Write one example of magmatic systems where liquid immiscibility occurs.
- (g) Consider a condition in which mantle convection is stopped. What could be the primary cause?
- (h) If the bulk composition of magma is $An_{70}Ab_{30}$, then what will be the composition of the final crystallization product in equilibrium crystallization and fractional crystallization?
- (i) In the system diopside-anorthite, the presence of phenocrysts would indicate that the temperatures couldn't be more than _____ above the eutectic.

- (j) Consider an alkaline rock having adequate or excess silica (silica saturated or oversaturated rock) but deficient in alumina. What will be its mineralogy?
- (k) $\text{MgAl}_2\text{O}_4 + \text{MgSiO}_3 = \text{Mg}_2\text{SiO}_4 + \text{Mg}_3\text{Al}_3\text{Si}_3\text{O}_{12}$. In which depth of mantle does this reaction happen?
- (l) Explain the following terms: Picrite, Uralitization.
- (m) Identify the closed crystal form shown below:

